



DO-IT-YOURSELF: PROJECT

operate using three 30-watt power solar panels connected in series. Each panel should have a maximum power voltage (Vmp) of about 17.5 to 18 volts. Ensure that the PV panel output is connected to the +Vdc and -Vdc terminals, and make sure to disconnect the AC-DC converter output.

You can also set up a battery backup using four 12V lead-acid or lithium-ion batteries connected in series. Connect the battery output to the +Vdc and -Vdc terminals but be sure to disconnect the AC-DC converter output connections before doing so. These batteries, while expensive upfront, should have a longer lifespan since they will be used only occasionally.

Electrical safety. The heater coil operates on 48V DC, which is considered a 'safe' voltage. When selecting an AC-DC converter, ensure it has mains supply isolation,

typically achieved with an internal transformer (forward or flyback type). This design prevents the 230V AC mains from appearing at the converter's output, if it malfunctions. Always use a 3-pin socket and ensure proper grounding for added safety.

Thermal safety. To prevent an unlikely circuit failure where continuous power to the heater could cause its temperature to exceed 150°C, it is advisable to include a thermal fuse in series with the coil. This fuse is mounted on the heater coil, similar to how the thermistor is placed. If the heater's temperature surpasses 150°C, the thermal fuse will melt, cutting off the power supply to the coil. Please note that a melted thermal fuse should be replaced with a new one.

In the circuit diagram, the thermal fuse is not shown. In the ab-

sence of a thermal fuse, users should ensure that they turn off the switch once the cooking process is complete to prevent overheating.

This slow cooker is designed to preserve the nutrients and flavours of millets. It utilises common heater coils and casseroles, making it straightforward to assemble and operate. It contributes to electricity savings and can run on solar power in bright sunlight. If necessary, it can be linked to a home inverter or equipped with a battery backup. Widespread adoption of this appliance can enhance people's health, because it is common knowledge that millets are bullets of nutrients. **EFY**

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